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Introduction

Acute respiratory failure (ARF) requiring intensive care unit (ICU) admission is a leading cause of critical illness. Although best practices for mechanical ventilation are well established, adherence to optimal strategies remains inconsistent because patient physiology and ventilator settings evolve non-linearly and require hundreds of time-sensitive decisions. Reinforcement learning (RL) is a sequential-decision framework that can be used to learn personalized treatment policies from large observational datasets.

Objective

Aim. Train and externally validate a deep reinforcement learning agent for personalized hourly ventilator management in acute respiratory failure.

Hypothesis. Hours in which clinician actions agree with the agent's recommendation are associated with lower in-hospital mortality, after adjustment for age and baseline severity.

Methods

- Cohort.** Adults receiving invasive mechanical ventilation for ≥ 4 hours, drawn from the Common Longitudinal ICU Format (CLIF) version of MIMIC-IV for training, and three independent CLIF-standardized ICU datasets for external validation.
- Agent.** Double deep Q-network (DQN) with 1-hour decision intervals.
- State space.** Most recent vitals, labs, ventilator settings, and continuous medications.
- Action space.** 2×2 grid:
 - Ventilator mode: controlled vs. uncontrolled
 - Oxygenation support: low vs. high PEEP/FiO₂
- Reward.** Captures hourly change in arterial pH and PaO₂/FiO₂ ratio, plus terminal survival.
- Concordance validation.** For each site, we estimated the adjusted odds ratio of in-hospital mortality per 10 percentage-point increase in physician-agent concordance, adjusting for age at admission and baseline illness severity.

Acknowledgements

- NHLBI K08HL150291
- NLM R01LM014263

CLIF Information



Figure 1: Study Design and RL Workflow

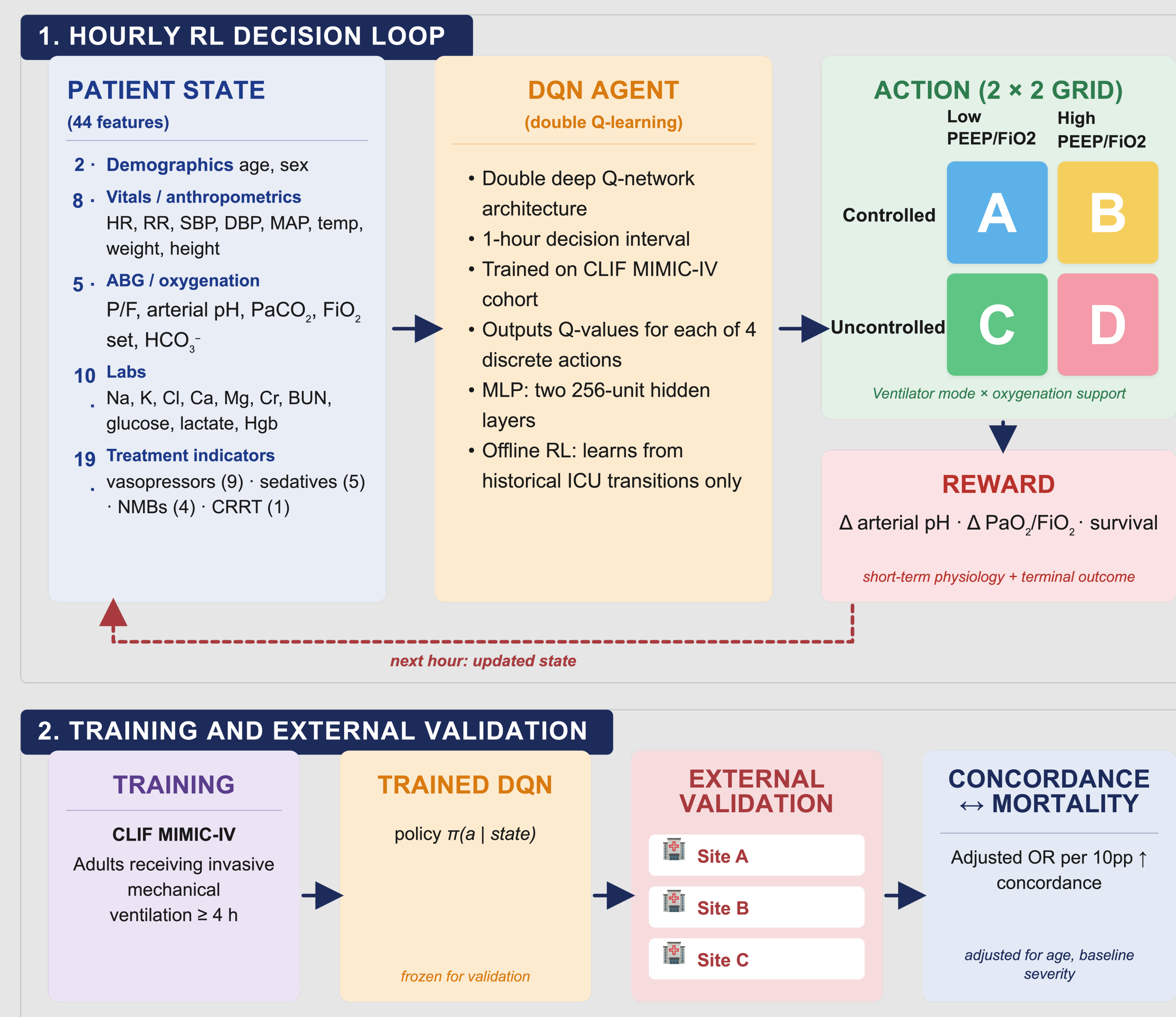


Figure 2: Ventilator Settings Distribution by CLIF Sites (RL Agent vs



Results

In external validation across three CLIF-standardized ICU datasets, higher physician-agent concordance was associated with lower confounder-adjusted odds of in-hospital mortality:

- Site A:** -3.2% per 10pp $\uparrow$ concordance (95% CI, 0.9–5.5%; $p = 0.006$)
- Site B:** -4.8% (95% CI, 2.6–7.0%; $p < 0.001$)
- Site C:** -8.4% (95% CI, 5.7–11.0%; $p < 0.001$)

The RL agent more frequently selected uncontrolled ventilation modes, reserving controlled modes for patients receiving higher sedation doses or with severe gas-exchange impairment (**Figure 2**). Representative physician and agent action sequences over time are shown in **Figure 3**.

Conclusion

- Main finding:** Across three independent CLIF-standardized ICU databases, every 10 percentage-point increase in clinician-RL agent concordance was associated with 3 – 8% lower adjusted odds of in-hospital mortality (all $p \leq 0.006$).
- Replication across heterogeneous sites:** with different case mix, protocols, and EHR systems, the study supports generalization beyond the institution where the agent was trained.
- Methodological contribution:** CLIF made it possible to deploy a single trained policy and a single evaluation pipeline at multiple ICUs without sharing patient-level data. A scalable template for federated critical-care AI research.
- Next steps:** Richer action spaces (continuous PEEP/FiO₂, tidal volume), additional clinical features and reward components (e.g., sedation depth), and prospective shadow-mode evaluation in a live ICU.

Figure 3: Representative patient timelines for physician and RL agent actions

